

Filière : Ingénierie des systèmes d'information et big data

Rapport des TPs

Ingénierie et maquette numériques

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TP1
Gestion de versions
avec Git & GitHub

1. Initialisation du dépôt

Le TP commence par la création et l'initialisation du dépôt Git local dans le dossier git-exercices. Après plusieurs tentatives d'accès au bon répertoire, la commande git init a été exécutée avec succès.

Commandes utilisées :

```
cd "C:\Users\x\OneDrive\Desktop\git-exercices"
```

```
git init
```

```
git status
```

Résultat observé : Git a été initialisé et les dossiers css/, html/ et js/ sont apparus comme fichiers non suivis (untracked files).

2. Premier commit — Initial commit

Les trois fichiers du projet ont été ajoutés à la zone de staging puis validés dans un commit initial.

Commandes utilisées :

```
git add *
```

```
git commit -m "Initial commit"
```

Les fichiers suivants ont été créés et committés :

- css/style.css — feuille de styles avec body { background: blue; }
- html/index.html — page HTML de base avec liens vers CSS et JS
- js/index.js — fichier JavaScript avec console.log("js is ready")

Hash du commit : 218ecb6 (root-commit)

```

C:\Users\x\OneDrive\Desktop\git-exercices>git add *
C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   css/style.css
        new file:   html/index.html
        new file:   js/index.js

C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "Initial commit"
[master (root-commit) 218ecb6] Initial commit
 3 files changed, 18 insertions(+)
 create mode 100644 css/style.css
 create mode 100644 html/index.html
 create mode 100644 js/index.js

C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch master
nothing to commit, working tree clean

C:\Users\x\OneDrive\Desktop\git-exercices>git branch dev
C:\Users\x\OneDrive\Desktop\git-exercices>git branch
  dev
* master

C:\Users\x\OneDrive\Desktop\git-exercices>git checkout dev
Switched to branch 'dev'

C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch dev
nothing to commit, working tree clean

C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch dev
nothing to commit, working tree clean

```

3. Création et utilisation de la branche dev

Une nouvelle branche dev a été créée à partir de master pour isoler les développements.

Commandes utilisées :

`git branch dev`

`git branch` → affiche : `* master / dev`

`git checkout dev` → Switched to branch 'dev'

Sur la branche dev, le fichier `css/style.css` a été modifié pour changer la couleur de fond :

```
body { background: red; }
```

Le changement a été stagé et commité :

`git add css/style.css`

`git commit -m "[CSS] changed the body background color"`

Hash du commit : `a649b9e`. Un `git push` simple a retourné une erreur (No configured push destination) car aucun remote n'était encore configuré.

4. Merge et résolution de conflit (master ↔ dev)

De retour sur master, le fichier style.css a aussi été modifié puis commité ([CSS] rechanged the body background color — hash ab6892a), créant ainsi une divergence avec la branche dev.

La fusion avec git merge dev a généré un conflit de contenu sur css/style.css :

CONFLICT (content): Merge conflict in css/style.css

Le conflit a été résolu manuellement puis commité :

```
git add css/style.css
```

```
git commit -m "Resolved merge conflict between master and dev" .
```

6. Simulation d'une branche master — branch-master

Pour simuler le workflow de fusion sans altérer master, une branche branch-master a été créée depuis dev :

```
git checkout dev
```

```
git checkout -b branch-master
```

branch-a a d'abord été mergée dans branch-master (fast-forward sans conflit) :

```
git merge branch-a → Fast-forward, 1 file changed
```

La fusion de branch-b a ensuite provoqué un conflit sur readme.md :

CONFLICT (content): Merge conflict in readme.md

Le conflit a été résolu en conservant les deux jeux d'instructions dans le fichier, puis commité :

```
git commit -m "making peace between branch-a and branch-b" → hash e32e9a1
```

7. Comparaisons entre branches

Plusieurs comparaisons ont été effectuées pour valider l'état des branches :

- `git diff dev master` → aucune différence sur les fichiers CSS
- `git diff branch-master dev` → readme.md absent de dev, présent dans branch-master
- `git diff master branch-master` → readme.md apparaît comme nouveau fichier dans branch-master

8. Configuration du remote et push vers GitHub

Un dépôt GitHub a été créé à l'adresse <https://github.com/assmaa-aza/git-exercices.git>, puis configuré comme remote :

```
git remote add origin https://github.com/assmaa-aza/git-exercices.git
```

```
git remote -v → origin fetch & push confirmés
```

Toutes **les branches ont été poussées** en une seule commande :

```
git push -u origin --all
```

9. Récapitulatif des commits

Hash	Branche	Message
218ecb6	master	Initial commit
a649b9e	dev	[CSS] changed the body background color
ab6892a	master	[CSS] rechanged the body background color
56f4bd1	master	Resolved merge conflict between master and dev
5552b60	newbranch	added a readme.md file
985480a	branch-a	[Readme] Added branch-a instructions
1aef1c0	branch-b	[Readme] Added branch-b instructions
e32e9a1	branch-master	making peace between branch-a and branch-b

9. Captures d'écrans des commandes :

```
C:\Users\x>cd desktop
C:\Users\x\Desktop>git status
fatal: not a git repository (or any of the parent directories): .git
C:\Users\x\Desktop>cd git-exercices
The system cannot find the path specified.
C:\Users\x\Desktop>cd git-exercices
The system cannot find the path specified.
C:\Users\x\Desktop>cd "C:\Users\x\OneDrive\Desktop\git-exercices"
C:\Users\x\OneDrive\Desktop\git-exercices>git status
fatal: not a git repository (or any of the parent directories): .git
C:\Users\x\OneDrive\Desktop\git-exercices>status
'status' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\x\OneDrive\Desktop\git-exercices>git status
fatal: not a git repository (or any of the parent directories): .git
C:\Users\x\OneDrive\Desktop\git-exercices>git status
fatal: not a git repository (or any of the parent directories): .git
C:\Users\x\OneDrive\Desktop\git-exercices>git status
fatal: not a git repository (or any of the parent directories): .git
C:\Users\x\OneDrive\Desktop\git-exercices>git status
fatal: not a git repository (or any of the parent directories): .git
C:\Users\x\OneDrive\Desktop\git-exercices>git init
Initialized empty Git repository in C:/Users/x/OneDrive/Desktop/git-exercices/.git/
C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    css/
    html/
    js/
nothing added to commit but untracked files present (use "git add" to track)
```

```
C:\Users\x\OneDrive\Desktop\git-exercices>git add *
C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch master
No commits yet
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   css/style.css
    new file:   html/index.html
    new file:   js/index.js
C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "Initial commit"
[master (root-commit) 218ecb6] Initial commit
 3 files changed, 18 insertions(+)
 create mode 100644 css/style.css
 create mode 100644 html/index.html
 create mode 100644 js/index.js
C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch master
nothing to commit, working tree clean
C:\Users\x\OneDrive\Desktop\git-exercices>git branch dev
C:\Users\x\OneDrive\Desktop\git-exercices>git branch
  dev
* master
C:\Users\x\OneDrive\Desktop\git-exercices>git checkout dev
Switched to branch 'dev'
C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch dev
nothing to commit, working tree clean
C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch dev
nothing to commit, working tree clean
```

```

C:\Users\x\OneDrive\Desktop\git-exercices>git branch dev
C:\Users\x\OneDrive\Desktop\git-exercices>git branch
* master
C:\Users\x\OneDrive\Desktop\git-exercices>git checkout dev
Switched to branch 'dev'
C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch dev
nothing to commit, working tree clean
C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch dev
nothing to commit, working tree clean
C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch dev
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   css/style.css

no changes added to commit (use "git add" and/or "git commit -a")
C:\Users\x\OneDrive\Desktop\git-exercices>git add css/style.css
C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "[CSS] changed the body background color"
[dev a649b9e] [CSS] changed the body background color
1 file changed, 3 insertions(+)
C:\Users\x\OneDrive\Desktop\git-exercices>git push
fatal: No configured push destination.
Either specify the URL from the command-line or configure a remote repository using

    git remote add <name> <url>

and then push using the remote name

    git push <name>

C:\Users\x\OneDrive\Desktop\git-exercices>git checkout master
Switched to branch 'master'
C:\Users\x\OneDrive\Desktop\git-exercices>git add css/style.css

```

```

C:\Users\x\OneDrive\Desktop\git-exercices>git status commit -m "[CSS] rechanged the body background color"
error: unknown switch -m
usage: git status [<options>] [--] [<pathspec>...]

    -v, --[no-]verbose          be verbose
    -s, --[no-]short            show status concisely
    -b, --[no-]branch          show branch information
    --[no-]show-stash           show stash information
    --[no-]ahead-behind        compute full ahead/behind values
    --[no-]porcelain[=<version>]
                                machine-readable output
    --[no-]long                show status in long format (default)
    -z, --[no-]null            terminate entries with NUL
    -u, --[no-]untracked-files[=<mode>]
                                show untracked files, optional modes: all, normal, no. (Default: all)
    --[no-]ignored[=<mode>]
                                show ignored files, optional modes: traditional, matching, no. (Default: traditional)
    --[no-]ignore-submodules[=<when>]
                                ignore changes to submodules, optional when: all, dirty, untracked. (Default: all)
    --[no-]column[=<style>]
                                list untracked files in columns
    --no-renames               do not detect renames
    --renames                  opposite of --no-renames
    -M, --find-renames[=<n>]
                                detect renames, optionally set similarity index

C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "[CSS] rechanged the body background color"
[master ab6892a] [CSS] rechanged the body background color
1 file changed, 3 insertions(+)
C:\Users\x\OneDrive\Desktop\git-exercices>git checkout master
Already on 'master'
C:\Users\x\OneDrive\Desktop\git-exercices>git pull
There is no tracking information for the current branch.
Please specify which branch you want to merge with.
See git-pull(1) for details.

    git pull <remote> <branch>

If you wish to set tracking information for this branch you can do so with:

    git branch --set-upstream-to=<remote>/<branch> master

```



```

C:\Users\x\OneDrive\Desktop\git-exercises>git merge dev
Auto-merging css/style.css
CONFLICT (content): Merge conflict in css/style.css
Automatic merge failed; fix conflicts and then commit the result.

C:\Users\x\OneDrive\Desktop\git-exercises>git add css/style.css

C:\Users\x\OneDrive\Desktop\git-exercises>git commit -m "Resolved merge conflict between master and dev"
[master 56f4bd1] Resolved merge conflict between master and dev

C:\Users\x\OneDrive\Desktop\git-exercises>git pull
There is no tracking information for the current branch.
Please specify which branch you want to merge with.
See git-pull(1) for details.

    git pull <remote> <branch>

If you wish to set tracking information for this branch you can do so with:

    git branch --set-upstream-to=<remote>/<branch> master

C:\Users\x\OneDrive\Desktop\git-exercises>git merge dev
Already up to date.

C:\Users\x\OneDrive\Desktop\git-exercises>git checkout master
Already on 'master'

C:\Users\x\OneDrive\Desktop\git-exercises>git pull
There is no tracking information for the current branch.
Please specify which branch you want to merge with.
See git-pull(1) for details.

    git pull <remote> <branch>

If you wish to set tracking information for this branch you can do so with:

    git branch --set-upstream-to=<remote>/<branch> master

C:\Users\x\OneDrive\Desktop\git-exercises>git merge dev
Already up to date.

C:\Users\x\OneDrive\Desktop\git-exercises>touch readme.md
'touch' is not recognized as an internal or external command,
operable program or batch file.

```

```

C:\Users\x\OneDrive\Desktop\git-exercises>git diff branch-a branch-b

C:\Users\x\OneDrive\Desktop\git-exercises>git log branch-a
commit 5552b606f2beaa26007274612b9e2fdc8e04f882 (HEAD -> newbranch, branch-b, branch-a)
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:48:00 2026 +0000

    added a readme.md file

commit 56f4bd1f7b60515eae484a8c72016c6b31e0f4d (master)
Merge: ab6892a a649b9e
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:37:47 2026 +0000

    Resolved merge conflict between master and dev

commit ab6892a5fe2516b8cf76df05a1084db16e2605c5
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:34:13 2026 +0000

    [CSS] rechanged the body background color

commit a649b9eb68bc67b3000840cf931c49e1e2ab73fa (dev)
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:31:20 2026 +0000

    [CSS] changed the body background color

commit 218ecbb645796e1989d0ef93a1a6ae65cc3d9cc6
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:20:11 2026 +0000

    Initial commit

C:\Users\x\OneDrive\Desktop\git-exercises>git log branch-b
commit 5552b606f2beaa26007274612b9e2fdc8e04f882 (HEAD -> newbranch, branch-b, branch-a)
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:48:00 2026 +0000

    added a readme.md file

commit 56f4bd1f7b60515eae484a8c72016c6b31e0f4d (master)
Merge: ab6892a a649b9e
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:37:47 2026 +0000

    Resolved merge conflict between master and dev

commit ab6892a5fe2516b8cf76df05a1084db16e2605c5

```

```

Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:34:13 2026 +0000

    [CSS] rechanged the body background color

commit a649b9eb68bc67b3000840cf931c49e1e2ab73fa (dev)
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:31:20 2026 +0000

    [CSS] changed the body background color

commit 218ecbb645796e1989d0ef93a1a6ae65cc3d9cc6
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:20:11 2026 +0000

    Initial commit

C:\Users\x\OneDrive\Desktop\git-exercices>git checkout branch-a
Switched to branch 'branch-a'

C:\Users\x\OneDrive\Desktop\git-exercices>git branch
* branch-a
  branch-b
  dev
  master
  newbranch

C:\Users\x\OneDrive\Desktop\git-exercices>git add readme.md

C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "[Readme] Added branch-a instructions"
[branch-a 985480a] [Readme] Added branch-a instructions
1 file changed, 1 insertion(+)

C:\Users\x\OneDrive\Desktop\git-exercices>git checkout branch-b
Switched to branch 'branch-b'

C:\Users\x\OneDrive\Desktop\git-exercices>git add readme.md
'd' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\x\OneDrive\Desktop\git-exercices>git add readme.md

C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "[Readme] Added branch-b instructions"
[branch-b 1aef1c0] [Readme] Added branch-b instructions
1 file changed, 1 insertion(+)

C:\Users\x\OneDrive\Desktop\git-exercices>git diff branch-a branch-b
diff --git a/readme.md b/readme.md
index 47645ed..ca13140 100644
--- a/readme.md

```

```

C:\Users\x\OneDrive\Desktop\git-exercices>git diff branch-a branch-b
diff --git a/readme.md b/readme.md
index 47645ed..ca13140 100644
--- a/readme.md
+++ b/readme.md
@@ -1,1 @@
-Branch-a instructions
\ No newline at end of file
+Branch-b instructions
\ No newline at end of file

C:\Users\x\OneDrive\Desktop\git-exercices>git log branch-a
commit 985480a81e6a167dbbccc65e35e74f988e37facfc (branch-a)
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:53:27 2026 +0000

    [Readme] Added branch-a instructions

commit 5552b606f2beaa26007274612b9e2fdc8e04f882 (newbranch)
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:48:00 2026 +0000

    added a readme.md file

commit 56f4bd1f7b60515eae6484a8c72016c6b31e0f4d (master)
Merge: ab6892a a649b9e
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:37:47 2026 +0000

    Resolved merge conflict between master and dev

commit ab6892a5fe2516b8cf76df05a1084db16e2605c5
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:34:13 2026 +0000

    [CSS] rechanged the body background color

commit a649b9eb68bc67b3000840cf931c49e1e2ab73fa (dev)
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:31:20 2026 +0000

    [CSS] changed the body background color

commit 218ecbb645796e1989d0ef93a1a6ae65cc3d9cc6
Author: asma <assmaaazaroual@gmail.com>
Date:   Wed Feb 18 11:20:11 2026 +0000

    Initial commit

```

```

C:\Users\x\OneDrive\Desktop\git-exercices>git log branch-b
commit 1aef1c04f18fd328ec692f19fd67e12990ede59a (HEAD -> branch-b)
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:54:36 2026 +0000

    [Readme] Added branch-b instructions

commit 5552b606f2beaa26007274612b9e2fdc8e04f882 (newbranch)
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:48:00 2026 +0000

    added a readme.md file

commit 56f4bd1f7b60515eae6484a8c72016c6b31e0f4d (master)
Merge: ab6892a a649b9e
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:37:47 2026 +0000

    Resolved merge conflict between master and dev

commit ab6892a5fe2516b8cf76df05a1084db16e2605c5
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:34:13 2026 +0000

    [CSS] rechanged the body background color

commit a649b9eb68bc67b3000840cf931c49e1e2ab73fa (dev)
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:31:20 2026 +0000

    [CSS] changed the body background color

commit 218ecbb645796e1989d0ef93a1a6ae65cc3d9cc6
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:20:11 2026 +0000

    Initial commit

C:\Users\x\OneDrive\Desktop\git-exercices>git checkout dev
Switched to branch 'dev'

C:\Users\x\OneDrive\Desktop\git-exercices>git checkout -b branch-master
Switched to a new branch 'branch-master'

C:\Users\x\OneDrive\Desktop\git-exercices>git merge branch-a
Updating a649b9e..985480a
Fast-forward
 readme.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 readme.md

```

```

C:\Users\x\OneDrive\Desktop\git-exercices>git status
On branch branch-master
All conflicts fixed but you are still merging.
(use "git commit" to conclude merge)

Changes to be committed:
  modified:   readme.md

C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "making peace between branch-a and branch-b"
[branch-master e32e9a1] making peace between branch-a and branch-b

C:\Users\x\OneDrive\Desktop\git-exercices>git log
commit e32e9a13e7e641a2edc9a8727ef42bc9a5e50029 (HEAD -> branch-master)
Merge: 985480a 1aef1c0
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 12:00:37 2026 +0000

    making peace between branch-a and branch-b

commit 1aef1c04f18fd328ec692f19fd67e12990ede59a (branch-b)
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:54:36 2026 +0000

    [Readme] Added branch-b instructions

commit 985480a81e6a167dbcc65e35e74f988e37facfc (branch-a)
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:53:27 2026 +0000

    [Readme] Added branch-a instructions

commit 5552b606f2beaa26007274612b9e2fdc8e04f882 (newbranch)
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:48:00 2026 +0000

    added a readme.md file

commit 56f4bd1f7b60515eae6484a8c72016c6b31e0f4d (master)
Merge: ab6892a a649b9e
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:37:47 2026 +0000

    Resolved merge conflict between master and dev

commit ab6892a5fe2516b8cf76df05a1084db16e2605c5
Author: asma <assmaaazaroual@gmail.com>
Date: Wed Feb 18 11:34:13 2026 +0000

```

```

    [CSS] rechanged the body background color
commit a649b9eb68bc67b3000840cf931c49e1e2ab73fa (dev)
Author: asma <assmaazaroual@gmail.com>
Date:   Wed Feb 18 11:31:20 2026 +0000

    [CSS] changed the body background color
commit 218ecbb645796e1989d0ef931a6ae65cc3d9cc6
Author: asma <assmaazaroual@gmail.com>
Date:   Wed Feb 18 11:20:11 2026 +0000

    Initial commit
C:\Users\x\OneDrive\Desktop\git-exercices>git diff dev master
C:\Users\x\OneDrive\Desktop\git-exercices>git diff branch-master dev
diff --git a/readme.md b/readme.md
deleted file mode 100644
index 084e95a..0000000
--- a/readme.md
+++ /dev/null
@@ -1,3 +0,0 @@
-//Instructions goes here...
-Branch-a instructions...
-Branch-b instructions...

C:\Users\x\OneDrive\Desktop\git-exercices>git diff master branch-master
diff --git a/readme.md b/readme.md
new file mode 100644
index 0000000..084e95a
--- /dev/null
+++ b/readme.md
@@ -0,0 +1,3 @@
+//Instructions goes here...
+Branch-a instructions...
+Branch-b instructions...

C:\Users\x\OneDrive\Desktop\git-exercices>git push -all
error: did you mean "--all" (with two dashes)?

C:\Users\x\OneDrive\Desktop\git-exercices>git push --all
fatal: No configured push destination.
Either specify the URL from the command-line or configure a remote repository using

    git remote add <name> <url>

and then push using the remote name

```

```

C:\Users\x\OneDrive\Desktop\git-exercices>git remote add origin https://github.com/assmaa-aza/git-exercices.git
C:\Users\x\OneDrive\Desktop\git-exercices>git remote -v
origin https://github.com/assmaa-aza/git-exercices.git (fetch)
origin https://github.com/assmaa-aza/git-exercices.git (push)

```

```

C:\Users\x\OneDrive\Desktop\git-exercices>git push -u origin --all
Enumerating objects: 28, done.
Counting objects: 100% (28/28), done.
Delta compression using up to 2 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (28/28), 2.40 KiB | 6.00 KiB/s, done.
Total 28 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), done.
To https://github.com/assmaa-aza/git-exercices.git
 * [new branch]      branch-a -> branch-a
 * [new branch]      branch-b -> branch-b
 * [new branch]      branch-master -> branch-master
 * [new branch]      dev -> dev
 * [new branch]      master -> master
 * [new branch]      newbranch -> newbranch
branch 'branch-a' set up to track 'origin/branch-a'.
branch 'branch-b' set up to track 'origin/branch-b'.
branch 'branch-master' set up to track 'origin/branch-master'.
branch 'dev' set up to track 'origin/dev'.
branch 'master' set up to track 'origin/master'.
branch 'newbranch' set up to track 'origin/newbranch'.

```

10. Conclusion

Ce TP a permis de mettre en pratique les commandes fondamentales de Git dans un contexte de développement web réel. Les principaux apprentissages sont :

- Initialisation d'un dépôt local et création d'un premier commit structuré.
- Gestion des branches (création, navigation, suppression) pour isoler les développements.
- Configuration d'un remote GitHub et push de l'ensemble des branches.

TP
Git – Branches
& Fusion

1. Introduction

Ce compte rendu présente la réalisation du TP Git portant sur la gestion des branches, les commits et la fusion (merge). L'objectif est de comprendre et de pratiquer les commandes fondamentales de Git pour travailler en parallèle sur plusieurs branches d'un même dépôt.

Le TP est structuré en trois exercices progressifs :

- **Exercice 1** : Création de la branche style, modification CSS, retour sur master, modification HTML, retour sur style et nouvelle modification CSS.
- **Exercice 2** : Fusion de la branche style dans master (git merge) et observation du résultat.
- **Exercice 3** : Création d'un dépôt avec conflits, tentative de fusion et résolution manuelle.

2. Exercice 1 – Branches et Commits

2.1 Création de la branche style et modification CSS

La première étape consiste à créer une branche nommée style depuis le dépôt existant, s'y placer, modifier le fichier style.css (changement de la couleur de fond), puis commiter les modifications.

Commandes utilisées :

```
git branch style
git checkout style
git add .
git commit -m "change background color to green"
```

```
C:\Users\x\OneDrive\Desktop\git-exercices>git branch style
C:\Users\x\OneDrive\Desktop\git-exercices>git checkout style
Switched to branch 'style'
C:\Users\x\OneDrive\Desktop\git-exercices>git add .
C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "change background color to green"
On branch style
nothing to commit, working tree clean
C:\Users\x\OneDrive\Desktop\git-exercices>git add .
C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "change background color to green"
[style 7d5fb12] change background color to green
1 file changed, 1 insertion(+), 1 deletion(-)
```

Fig. 1 – Création de la branche style, bascule et commit de la modification CSS

La première tentative de commit affiche 'nothing to commit, working tree clean' car le fichier n'était pas encore sauvegardé. Après sauvegarde dans VS Code, le second `git add .` et `git commit` enregistre bien la modification : 1 file changed, 1 insertion(+), 1 deletion(-).

Le fichier `style.css` dans la branche `style` contient alors la nouvelle couleur de fond :



Fig. 2 – Contenu de `style.css` dans VS Code (modification de la couleur de fond)

2.2 Retour sur master : disparition des changements de style

On revient sur la branche `master` avec `git checkout master`. Les modifications apportées au fichier `style.css` dans la branche `style` ont disparu : le fichier retrouve son état d'avant le commit de `style`. C'est le comportement normal de Git : chaque branche maintient son propre état.

`git checkout master`

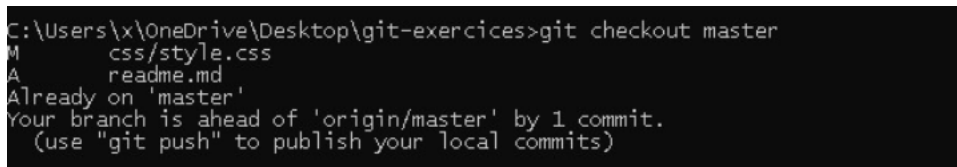


Fig. 3 – Retour sur master : 'Your branch is ahead of origin/master by 1 commit'

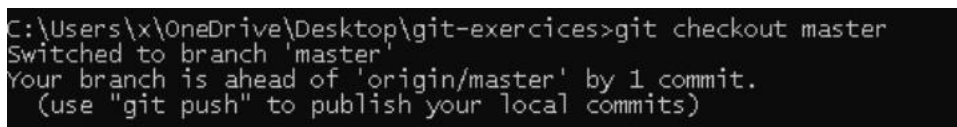


Fig. 4 – Confirmation du passage sur la branche master

2.3 Modification du fichier HTML dans master

Depuis la branche `master`, on modifie le fichier `index.html` pour ajouter un titre `h1`, puis on commite ces modifications.

```
git add .  
git commit -m "update html content"  
# → [master 37fc14e] update html content  
# → 1 file changed, 1 insertion(+)
```

```
<> index.html  
1  <!DOCTYPE html>  
2  <html class="no-js">  
3  <head>  
4      <meta charset="UTF-8">  
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" >  
6      <meta name="viewport" content="width=device-width, initial-scale=1<  
7      <meta name="description" content="">  
8      <title>Git exercices</title>  
9      <link rel="stylesheet" href="../css/style.css">  
10 </head>  
11 <body>  
12     <script src="../js/index.js" async defer></script>  
13     <h1>Welcome to my website</h1>  
14 </body>  
15 </html>
```

Fig. 5 – Contenu de index.html dans master avec ajout de la balise h1

Le fichier HTML dans master contient maintenant un titre h1 'Welcome to my website'. Ce commit n'existe pas dans la branche style.

2.4 Retour sur style : réapparition des modifications CSS

On repasse sur la branche style. Les modifications CSS (background: green) réapparaissent immédiatement, mais le contenu HTML ajouté dans master (le h1) a disparu. Chaque branche conserve strictement son propre historique.

git checkout style


```

index.html > html.no-js
1  <!DOCTYPE html>
2  <html class="no-js">
3  <head>
4      <meta charset="UTF-8">
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" >
6      <meta name="viewport" content="width=device-width, initial-scale=1
7      <meta name="description" content="">
8      <title>Git exercices</title>
9      <link rel="stylesheet" href="../../css/style.css">
10 </head>
11 <body>
12     <script src="../../js/index.js" async defer></script>
13 </body>
14 </html>

```

Fig. 6 – Retour sur style : les changements CSS ont réapparu, le h1 de master a disparu

```

# style.css
# style.css
1  body {
2  background: green;
3  }

```

Fig. 7 – style.css avec background: green réapparu dans la branche style

2.5 Seconde modification CSS (police et couleur noire)

Toujours dans la branche style, on effectue une nouvelle modification du fichier style.css : ajout de font-family: Arial et changement de la couleur de fond en noir. On commite ces changements.

```

git add .
git commit -m "change font and color to black"
# → [style e188652] change font and color to black
# → 1 file changed, 2 insertions(+), 1 deletion(-)

```

```
# style.css X
# style.css > body
1  body {}
2    background: black;
3    font-family: Arial;
4  }
```

Fig. 8 – style.css final : background black + font-family Arial

```
C:\Users\x\OneDrive\Desktop\git-exercices>git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

C:\Users\x\OneDrive\Desktop\git-exercices>git add.
git: 'add.' is not a git command. See 'git --help'.

The most similar command is
    add

C:\Users\x\OneDrive\Desktop\git-exercices>git add .
C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "update html content"
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

C:\Users\x\OneDrive\Desktop\git-exercices>git add .
C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "update html content"
[master 37fc14e] update html content
1 file changed, 1 insertion(+)

C:\Users\x\OneDrive\Desktop\git-exercices>git checkout style
Switched to branch 'style'

C:\Users\x\OneDrive\Desktop\git-exercices>git add .
C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "rechange style color to black "
On branch style
nothing to commit, working tree clean

C:\Users\x\OneDrive\Desktop\git-exercices>git add .
C:\Users\x\OneDrive\Desktop\git-exercices>git commit -m "change font and color to black"
[style e188652] change font and color to black
1 file changed, 2 insertions(+), 1 deletion(-)
```

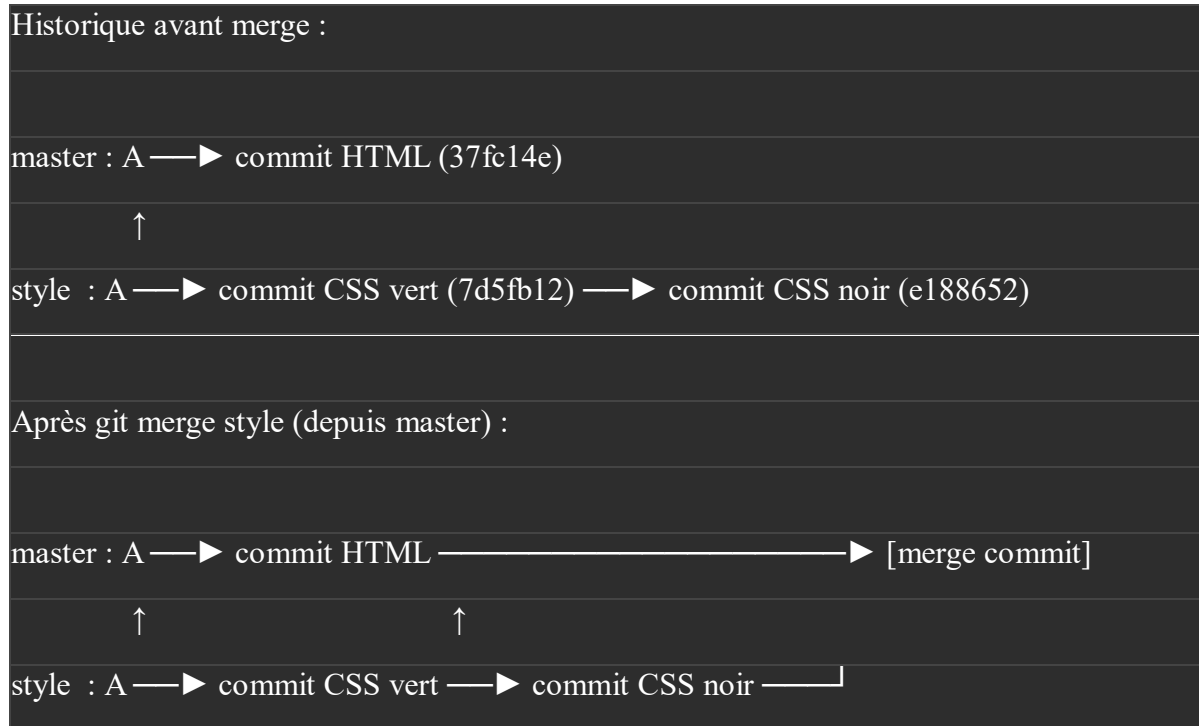
Fig. 9 – Vue complète des commandes de l'exercice 1 dans le terminal

3. Exercice 2 – Fusion des Branches (git merge)

L'objectif est de fusionner la branche style dans master afin d'obtenir un dépôt contenant à la fois les modifications HTML (master) et les modifications CSS (style).

3.1 Principe de la fusion avec commit

Les deux branches ont divergé : elles partagent un ancêtre commun mais ont chacune des commits supplémentaires différents. Git doit donc créer un nouveau commit de fusion (merge commit) ayant deux parents : le sommet de master et le sommet de style.



3.2 Exécution du merge

On se place dans la branche master, puis on lance git merge style. Comme les deux branches ont modifié des fichiers différents (index.html dans master, style.css dans style), Git fusionne automatiquement sans conflit.

```
git checkout master
git merge style
# -> Merge made by the 'ort' strategy.
# -> css/style.css | 2 ++-
# -> 1 file changed, 2 insertions(+), 1 deletion(-)
```

3.3 Résultat de la fusion

Après la fusion, on constate dans la copie de travail :

- index.html contient le h1 'Welcome to my website' ajouté dans master.
- style.css contient background: black et font-family: Arial de la branche style.
- L'historique (git log --oneline --graph) montre un commit de merge avec deux parents.

```
* [merge commit] Merge branch 'style'
|
| * e188652 change font and color to black
| * 7d5fb12 change background color to green
* | 37fc14e update html content
|/
* ... (commits précédents)
```

4. Exercice 3 – Gestion des Conflits

4.1 Principe

Un conflit Git survient lorsque deux branches modifient la même portion d'un même fichier texte. Git ne peut pas fusionner automatiquement et signale le conflit pour une résolution manuelle.

4.2 Mise en place du dépôt de test

On crée un nouveau dépôt contenant un fichier conflit.txt avec trois lignes de texte, puis on commite ce fichier initial.

```
git init depot-conflit
cd depot-conflit
# Créer conflit.txt avec :
# La première ligne
# La deuxième ligne
# La troisième ligne
git add .
git commit -m "commit initial"
```

4.3 Création de branches avec modifications conflictuelles

On crée deux branches qui modifient toutes les deux la première ligne du fichier de façon différente, ce qui créera un conflit à la fusion.

```
# Branche 1 : modifier ligne 1 en 'La première ligne modifiée'

git checkout -b branche1

git add .

git commit -m "branche1 : modifier ligne 1"

# Branche 2 : modifier ligne 1 en 'La première ligne changée'

git checkout master

git checkout -b branche2

git add .

git commit -m "branche2 : changer ligne 1"
```

4.4 Tentative de fusion et détection du conflit

On tente de fusionner branche1 dans branche2. Git détecte immédiatement le conflit sur la première ligne et interrompt la fusion :

```
git checkout branche2

git merge branche1

# → CONFLICT (content): Merge conflict in conflit.txt

# → Automatic merge failed; fix conflicts and then commit.
```

4.5 Contenu du fichier en conflit

Git modifie automatiquement le fichier conflit.txt pour faire apparaître les deux versions concurrentes entre des marqueurs spéciaux :

```
<<<<<<< HEAD
La première ligne changée
```

```
=====
La première ligne modifiée
>>>>>>> branche1
La deuxième ligne
La troisième ligne
```

- <<<<<<< HEAD : version de la branche courante (branche2).
- ===== : séparateur entre les deux versions en conflit.
- >>>>>>> branche1 : version de la branche source (branche1).
- Les lignes non conflictuelles (deuxième et troisième) sont automatiquement fusionnées.

4.6 Résolution et finalisation

Pour résoudre le conflit, on édite manuellement le fichier, on conserve la version souhaitée et on supprime tous les marqueurs Git, puis on finalise la fusion :

```
# Après édition manuelle de conflit.txt :
# La première ligne résolue
# La deuxième ligne
# La troisième ligne

git add conflit.txt
git commit -a
# → [branche2 ...] Merge branch 'branche1' into branche2

# En cas d'abandon :
git merge --abort
```

5. Conclusion

Ce TP a permis de maîtriser les concepts fondamentaux de la gestion de branches sous Git.

Les compétences développées comprennent :

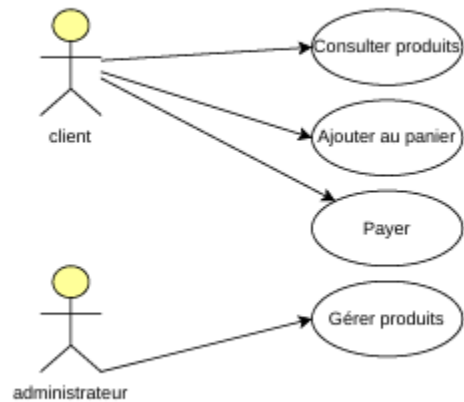
- La création et la navigation entre branches : git branch, git checkout.
- La compréhension de l'isolation des branches : chaque branche maintient son propre état de la copie de travail, indépendamment des autres.
- La réalisation de commits successifs sur des branches différentes : modifications CSS sur style, modifications HTML sur master.
- La fusion de deux branches divergentes (git merge) avec création automatique d'un merge commit à deux parents.
- La détection, la compréhension et la résolution manuelle de conflits lorsque deux branches modifient la même partie d'un fichier.

Le comportement observé lors des bascules entre branches illustre parfaitement le principe d'isolation de Git : la copie de travail reflète toujours l'état du dernier commit de la branche courante. La commande git merge permet d'unifier l'historique de deux branches en un seul commit, avec résolution automatique si les modifications portent sur des fichiers ou zones différents, et résolution manuelle obligatoire en cas de conflit.

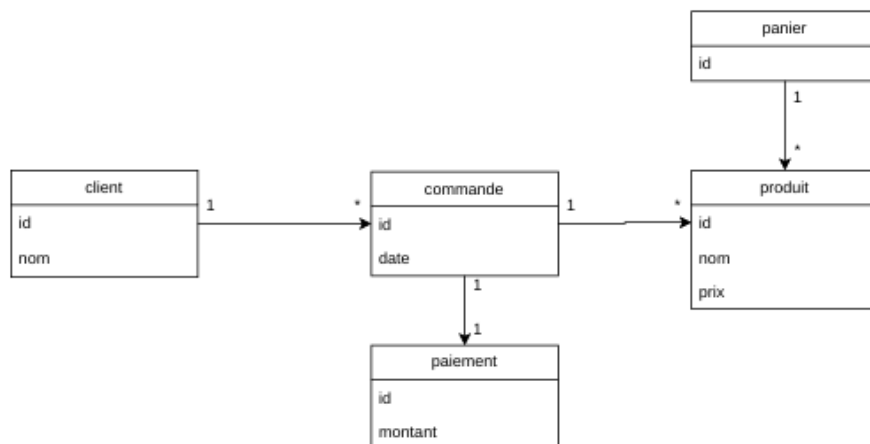
TP3
UML

Etude de cas 1 : Plateforme E-commerce

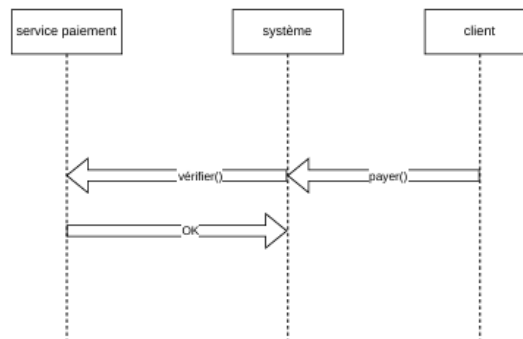
1.1 Diagramme cas d'utilisation



1.2 Diagramme de classes



1.3 Diagramme de séquence :



2.4.Workflow git

```
C:\Users\x>mkdir uml-tp
C:\Users\x>cd uml-tp
C:\Users\x\uml-tp>git init
Initialized empty Git repository in C:/Users/x/uml-tp/.git/
C:\Users\x\uml-tp>git remote add origin https://github.com/assmaa-aza/uml_1
C:\Users\x\uml-tp>git checkout -b main
Switched to a new branch 'main'
C:\Users\x\uml-tp>git checkout -b develop
Switched to a new branch 'develop'
C:\Users\x\uml-tp>git push -u origin develop
error: src refspec develop does not match any
error: failed to push some refs to 'https://github.com/assmaa-aza/uml_1'
C:\Users\x\uml-tp>echo "# UML TP Project" > README.md
C:\Users\x\uml-tp>git add .
C:\Users\x\uml-tp>git commit -m "initial commit"
[develop (root-commit) 31784a5] initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
C:\Users\x\uml-tp>git push -u origin develop
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 231 bytes | 21.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/assmaa-aza/uml_1
 * [new branch]      develop -> develop
branch 'develop' set up to track 'origin/develop'.
C:\Users\x\uml-tp>git checkout develop
Already on 'develop'
Your branch is up to date with 'origin/develop'.
C:\Users\x\uml-tp>git checkout develop
Already on 'develop'
Your branch is up to date with 'origin/develop'.
C:\Users\x\uml-tp>git checkout -b feature/paiement
Switched to a new branch 'feature/paiement'
C:\Users\x\uml-tp>git add .
```

```

C:\Users\x\uml-tp>git commit -m "ajout paiement"
On branch feature/paiement
nothing to commit, working tree clean

C:\Users\x\uml-tp>git branch
  develop
* feature/paiement

C:\Users\x\uml-tp>git checkout feature/paiement
Already on 'feature/paiement'

C:\Users\x\uml-tp>git push origin feature/paiement
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/paiement' on GitHub by visiting:
remote:   https://github.com/assmaa-aza/uml_1/pull/new/feature/paiement
remote:
To https://github.com/assmaa-aza/uml_1
 * [new branch]   feature/paiement -> feature/paiement

C:\Users\x\uml-tp>git checkout develop
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.

C:\Users\x\uml-tp>git merge feature/paiement
Already up to date.

C:\Users\x\uml-tp>git push origin develop
Everything up-to-date

C:\Users\x\uml-tp>git checkout main
error: pathspec 'main' did not match any file(s) known to git

C:\Users\x\uml-tp>git branch
* develop
  feature/paiement

C:\Users\x\uml-tp>git checkout -b main
Switched to a new branch 'main'

C:\Users\x\uml-tp>git checkout main
Already on 'main'

C:\Users\x\uml-tp>git merge develop
Already up to date.

C:\Users\x\uml-tp>git push origin main
fatal: unable to access 'https://github.com/assmaa-aza/uml_1/': Could not resolve host: github.com
C:\Users\x\uml-tp>git push origin main

```

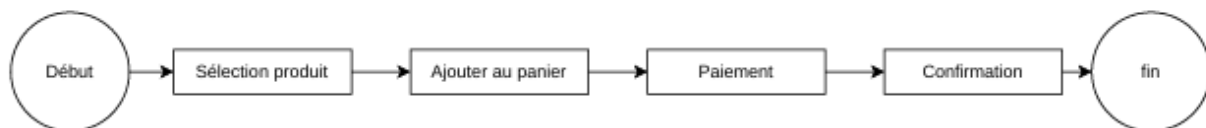
```

C:\Users\x\uml-tp>git push origin main
fatal: unable to access 'https://github.com/assmaa-aza/uml_1/': Could not resolve host: github.com

C:\Users\x\uml-tp>git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:   https://github.com/assmaa-aza/uml_1/pull/new/main
remote:
To https://github.com/assmaa-aza/uml_1
 * [new branch]   main -> main

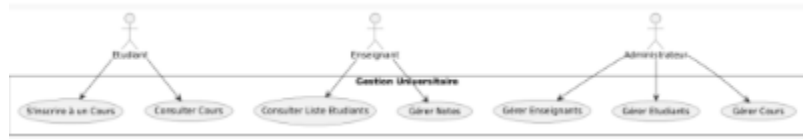
C:\Users\x\uml-tp>

```

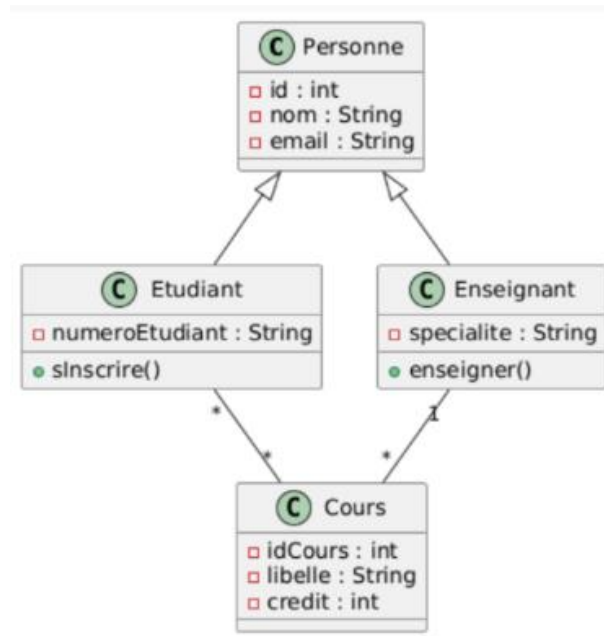


Etude Cas 2 : Application de gestion universitaire

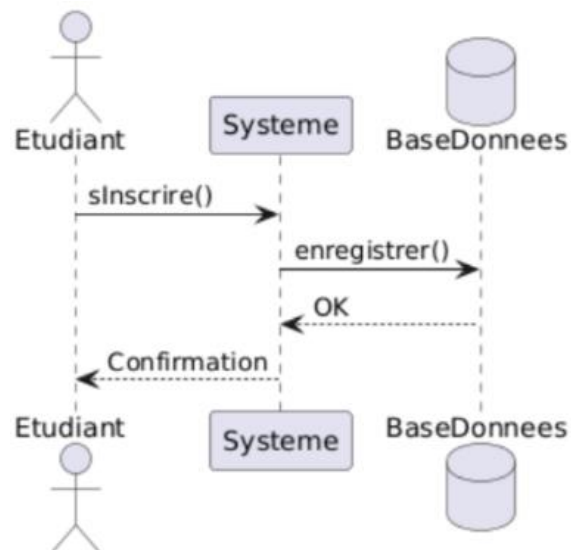
2.1 Diagramme Cas d'utilisation :



2.2 Diagramme de classe :



2.3 Diagramme de séquence :



2.4 workflow git :

```
C:\Users\x\uml-tp>git checkout -b feature/etudiant
Switched to a new branch 'feature/etudiant'

C:\Users\x\uml-tp>git checkout -b feature/cours
Switched to a new branch 'feature/cours'

C:\Users\x\uml-tp>git add .

C:\Users\x\uml-tp>git commit -m "ajout gestion etudiant"
On branch feature/cours
nothing to commit, working tree clean

C:\Users\x\uml-tp>git push origin feature/etudiant
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/etudiant' on GitHub by visiting:
remote:   https://github.com/assmaa-aza/uml_1/pull/new/feature/etudiant
remote:
To https://github.com/assmaa-aza/uml_1
 * [new branch]      feature/etudiant -> feature/etudiant
```

2.4.1 Créer un conflit volontaire

2.4.2 Créer les branches

`git checkout -b dev1`

`git checkout -b dev2`

```
C:\Users\x\uml-tp>git checkout -b dev1
Switched to a new branch 'dev1'

C:\Users\x\uml-tp>git checkout -b dev2
Switched to a new branch 'dev2'
```

2.4.3 Créer test.txt

2.4.3 Dans dev1 :

`git add .`

`git commit -m "modif dev1"`

```
C:\Users\x\uml-tp>git add .

C:\Users\x\uml-tp>git commit -m "modif dev1"
[dev2 0c8c1e8] modif dev1
1 file changed, 1 insertion(+)
create mode 100644 test.txt
```

2.4.4 Créer test2.txt

`git add .`

`git commit -m "modif dev2"`

```

C:\Users\x\uml-tp>git add .
C:\Users\x\uml-tp>git commit -m "modif dev2"
[dev2 4b7daed] modif dev2
1 file changed, 1 insertion(+)
create mode 100644 test2.txt
C:\Users\x\uml-tp>

```

2.4.5 Conflit

```

C:\Users\x\uml-tp>git merge dev1
Auto-merging test2.txt
CONFLICT (add/add): Merge conflict in test2.txt
Automatic merge failed; fix conflicts and then commit the result.

```

2.4.6 Le problème apparaît dans test2.txt

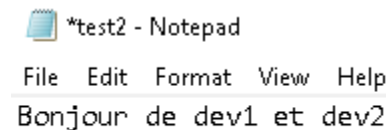


```

test2 - Notepad
File Edit Format View Help
|<<<<<<< HEAD
|Bonjour de dev2
|=====
|Bonjour dev2
|>>>>>>> dev1
|

```

2.4.7 Résolution en changeant le contenu



```

*test2 - Notepad
File Edit Format View Help
Bonjour de dev1 et dev2

```

```

C:\Users\x\uml-tp>git add .
C:\Users\x\uml-tp>git commit -m "résolution conflit"
[dev2 cd30bd0] résolution conflit
1 file changed, 1 insertion(+), 5 deletions(-)

```